

Multiple Choice (10 marks)

1. If $v(t) = 10 e^{-4t}u(t)$, and $V_{ab}(s)$ is related to $V(s)$ by $V_{ab}(s) = [\{3V(s) + 3s + 27\}/(s^2 + 6s + 8)]$, formulate $V_{ab}(t)$.
 - (a) $V_{ab}(t) = -15te^{-4t} - 15e^{-4t} + 18e^{-2t}$
 - (a) $V_{ab}(t) = 15te^{-4t} - 15e^{-4t} + 18e^{-2t}$
 - (a) $V_{ab}(t) = 30te^{-4t} + 30e^{-4t} - 36e^{-2t}$
 - (a) $V_{ab}(t) = 30te^{-4t} - 30e^{-4t} + 36e^{-2t}$
 - (a) $V_{ab}(t) = 15te^{-4t} + 15e^{-4t} - 18e^{-2t}$
2. Find the magnitude of E for a plane wave in free space, if the magnitude of H for this wave is 5 amp/meter.
 - (a) 1200 volts/meter
 - (b) 5 ohms
 - (c) 1883 volts/meter
 - (d) 940.5 volts/meter
 - (e) 5 volts/meter
3. In the toggle mode a JK flip-flop has
 - (a) $J = 0, K = 0$.
 - (b) $J = 1, K = 1$.
 - (c) $J = 0, K = 1$.
 - (d) $J = 1, K = 0$.
4. How many entries will be in the truth table of a 3 input NAND gate ?
 - (a) 6
 - (b) 12
 - (c) 4
 - (d) 10
 - (e) 18
5. Calculate the directivity of a short dipole.
 - (a) 3

- (b) 5
- (c) 2.5
- (d) 1
- (e) 2

True / False (10 marks)

Answer True or False

6. True or False: The gradient of the function f in cylindrical coordinates is $(z / ^3)u_{\phi} + (1/r)u_r - (1/r^3)u_z$.
6. True or False: Each hexadecimal digit is represented by exactly four binary bits.
7. True or False: An RC circuit has its lowest critical frequency at the origin, indicating it behaves similarly to an RL circuit.
8. True or False: When the length of a cable is doubled, its capacitance is also doubled, assuming other factors remain constant.
9. True or False: The sum $A + B$ equals $|375| + |475|$, and the difference $A - B$ equals $|-2 - 9 - 1| + |2 - 32|$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process of heating one mole of carbon dioxide at constant volume from 540 to 3540 F. Calculate the total heat added and explain the underlying principles involved.
12. Discuss the probability of obtaining exactly 2 heads and 1 tail when tossing a penny three times. Include the relevant calculations and implications for understanding probability in engineering contexts.

End of Paper